

Advanced (Habanero)

Q1. What is the remainder when 24 is divided by 5?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 0

Q2. Which number is divisible by 3 and 9?

- (A) 18
- (B) 20
- (C) 27
- (D) 30
- (E) 36

Q3. If $a = 12$ and $b = 4$, what is $a \div b$?

- (A) 2
- (B) 3
- (C) 4
- (D) 6
- (E) 8

Q4. What is the remainder when 37 is divided by 11?

- (A) 3
- (B) 4
- (C) 5
- (D) 6
- (E) 7

Q5. If a number is divisible by 5, it must end in which digit?

- (A) 0 or 5
- (B) 1 or 6
- (C) 2 or 7
- (D) 3 or 8
- (E) 4 or 9

Q6. What is $48 \div 6$?

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 10

Q7. If 18 is divided by 3, what is the quotient and remainder?

- (A) 6 R 0
- (B) 5 R 1
- (C) 6 R 1
- (D) 7 R 0
- (E) 7 R 1

Q8. What is $99 \div 9$?

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 11

Open Ended Questions (FRQ)

Q9. Perform the long division: $432 \div 12$. Show your work and find the quotient and remainder.

Q10. If $x = 24$ and $y = 5$, find the quotient and remainder when x is divided by y . Express the result as $q \cdot y + r$.

Chef's Tips

- Read each question carefully and choose the correct answer.
- For free response questions, show all work and simplify answers.
- Use a pencil and fill in the answer bubbles completely.